

Seat No: _____

Enrolment No: _____



School of Computer Applications
Department of Master of Computer Applications
MCA Semester-2 Summer – 2024 Exam

Subject Code: 140110201

Date: 26/02/2024

Subject Name: Data Structures (DS)

Duration: 02:30 Hours

Total Marks: 50

Instructions:

1. Attempt all the questions
2. Figures on the right indicate the marks.
3. Draw the figures and give suitable examples as and where necessary.

Q - 1 Answer The Following Questions.

- | | |
|--|---|
| 1. Define: Algorithm | 1 |
| 2. Define: Linear and Non-Linear Data Structures | 1 |
| 3. Explain primitive data types. | 1 |
| 4. What do you mean by algorithm analysis? How one can analyze it? | 3 |
| 5. Explain Asymptotic Notations in detail. | 4 |

OR

- | | |
|--|---|
| 5. Explain malloc() and free() functions in 'C'. Also discuss advantages of dynamic over static memory allocation. | 4 |
|--|---|

Q - 2 Answer The Following Questions.

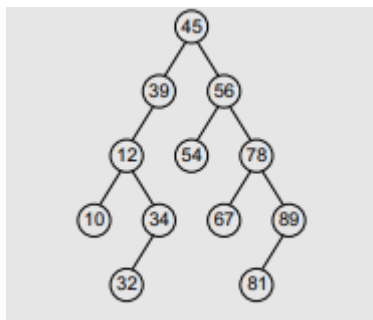
- | | |
|---|---|
| 6. List out applications of stack. | 1 |
| 7. Define: priority queue | 1 |
| 8. Compare linked list and a linear array. | 1 |
| 9. Explain the difference between a circular linked list and a singly linked list with suitable diagram. | 3 |
| 10. What is a circular queue? How do you check the queue full condition? Write an algorithm to Enqueue and Dequeue in a circular queue(Draw figure) | 4 |

OR

10. Convert the following infix expressions to their postfix equivalents using stack(Show every steps): $((A - B) + D / ((E + F) * G)$ 4

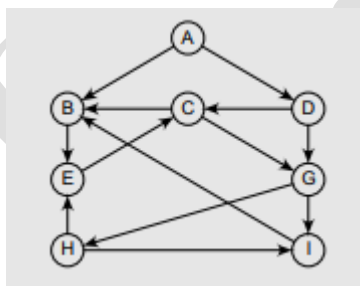
Q - 3 Answer The Following Questions.

11. Define: Spanning Tree. 1
12. Differentiate complete binary tree and full binary tree 1
13. Define: Directed Graph 1
14. Draw expression tree of given expression : $3+((5+9)*2$ 3
15. Consider the binary search tree given below. Now do the following operations: 4
- 1) Find the result of in-order, pre-order, and post-order traversals.
 - 2) Show the deletion of the root node
 - 3) Insert 11, 22, 33, 44, 55, 66, and 77 in the tree



OR

15. Consider the graph given below. Find out its depth-first and breadth-first traversal scheme.(Consider A as a starting vertex) 4



Q - 4 Answer The Following Questions.

16. Define a hash table 1
17. Which technique of searching an element in an array would you prefer to use and in which situation? 1
18. What is RL rotation in AVL tree. 1

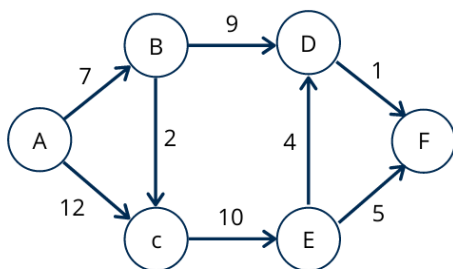
19. What is hashing? Explain any two techniques for collision resolution with appropriate example. 3
20. Search the number 50 from the given data using binary search technique. Illustrate the searching process. 10, 14, 20, 39, 41, 45, 49, 50, 60 4

OR

20. Explain Quick sort with example. 4
48,44,19,59,72,80,42,65,82,8,95,68

Q - 5 Answer The Following Questions.

21. Define doubly linked list. 1
22. What is Isolated vertex in Graph? 1
23. Define Degree of a node. 1
24. Create an AVL tree for the following : 21,26,30,9,4,14,28,18,15,10 3
25. Apply Dijkstra's algorithm to find shortest path on following graph with Node A as the starting node. 4



OR

25. Consider the graph given below. Find the minimum spanning tree of this graph using Prim's algorithm 4

